CHAPTER 7

Newton's Law of Gravitation



FIGURE 7.1 Johannes Kepler (left) showed how the planets move, and Isaac Newton (right) discovered that gravitational force caused them to move that way. ((left) unknown, Public Domain; (right) Sir Godfrey Kneller, Public Domain)

CHAPTER OUTLINE

7.1 Kepler's Laws of Planetary Motion

7.2 Newton's Law of Universal Gravitation and Einstein's Theory of General Relativity

INTRODUCTION What do a falling apple and the orbit of the moon have in common? You will learn in this chapter that each is caused by gravitational force. The motion of all celestial objects, in fact, is determined by the gravitational force, which depends on their mass and separation.

Johannes Kepler discovered three laws of planetary motion that all orbiting planets and moons follow. Years later, Isaac Newton found these laws useful in developing his law of universal gravitation. This law relates gravitational force to the masses of objects and the distance between them. Many years later still, Albert Einstein showed there was a little more to the gravitation story when he published his theory of general relativity.

7.1 Kepler's Laws of Planetary Motion

SECTION LEARNING OBJECTIVES

By the end of this section, you will be able to do the following:

- Explain Kepler’s three laws of planetary motion
- Apply Kepler’s laws to calculate characteristics of orbits

Section Key Terms

aphelion	Copernican model	eccentricity
Kepler’s laws of planetary motion	perihelion	Ptolemaic model

Concepts Related to Kepler's Laws of Planetary Motion

Examples of orbits abound. Hundreds of artificial satellites orbit Earth together with thousands of pieces of debris. The moon's orbit around Earth has intrigued humans from time immemorial. The orbits of planets, asteroids, meteors, and comets around the sun are no less interesting. If we look farther, we see almost unimaginable numbers of stars, galaxies, and other celestial objects orbiting one another and interacting through gravity.

All these motions are governed by gravitational force. The orbital motions of objects in our own solar system are simple enough to describe with a few fairly simple laws. The orbits of planets and moons satisfy the following two conditions:

- The mass of the orbiting object, m, is small compared to the mass of the object it orbits, M.
- The system is isolated from other massive objects.

Based on the motion of the planets about the sun, Kepler devised a set of three classical laws, called **Kepler's laws of planetary motion**, that describe the orbits of all bodies satisfying these two conditions:

1. The orbit of each planet around the sun is an ellipse with the sun at one focus.
2. Each planet moves so that an imaginary line drawn from the sun to the planet sweeps out equal areas in equal times.
3. The ratio of the squares of the periods of any two planets about the sun is equal to the ratio of the cubes of their average distances from the sun.

These descriptive laws are named for the German astronomer Johannes Kepler (1571–1630). He devised them after careful study (over some 20 years) of a large amount of meticulously recorded observations of planetary motion done by Tycho Brahe (1546–1601). Such careful collection and detailed recording of methods and data are hallmarks of good science. Data constitute the evidence from which new interpretations and meanings can be constructed. Let's look closer at each of these laws.

Kepler's First Law

The orbit of each planet about the sun is an ellipse with the sun at one focus, as shown in Figure 7.2. The planet's closest approach to the sun is called **perihelion** and its farthest distance from the sun is called **aphelion**.

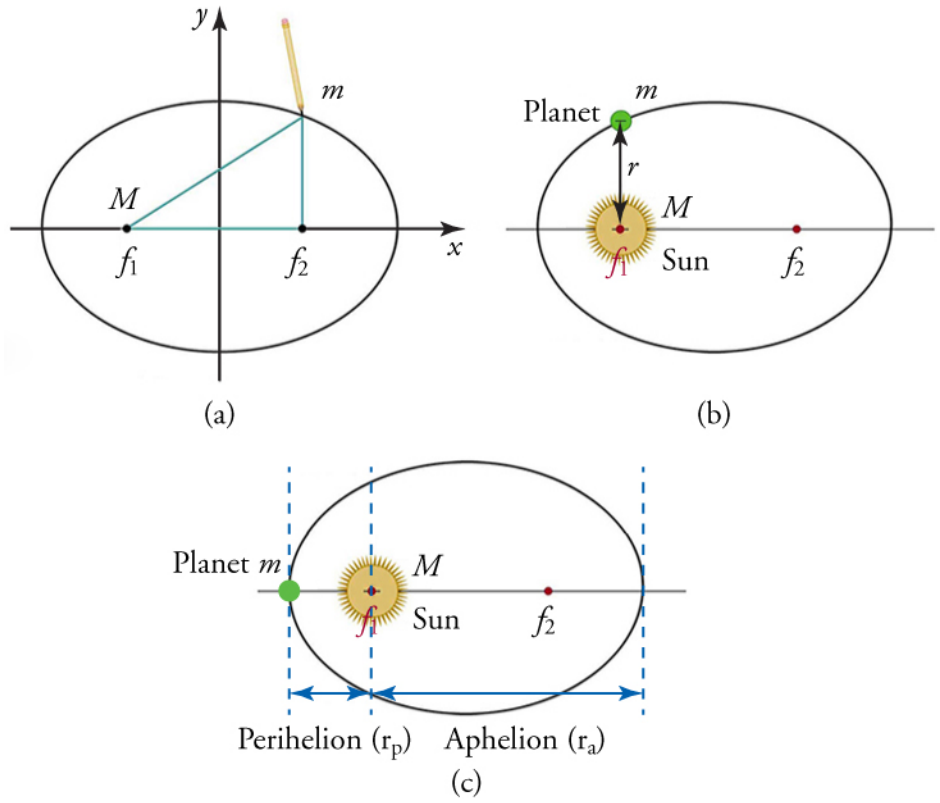


FIGURE 7.2 (a) An ellipse is a closed curve such that the sum of the distances from a point on the curve to the two foci (f_1 and f_2) is constant. (b) For any closed orbit, *m* follows an elliptical path with *M* at one focus. (c) The aphelion (*r*a) is the furthest distance between the planet and the sun, while the perihelion (*r*p) is the closest distance from the sun.

If you know the aphelion (r_a) and perihelion (r_p) distances, then you can calculate the semi-major axis (*a*) and semi-minor axis (*b*).

$$a = \frac{(r_a + r_p)}{2}$$
$$b = \sqrt{r_a r_p}$$

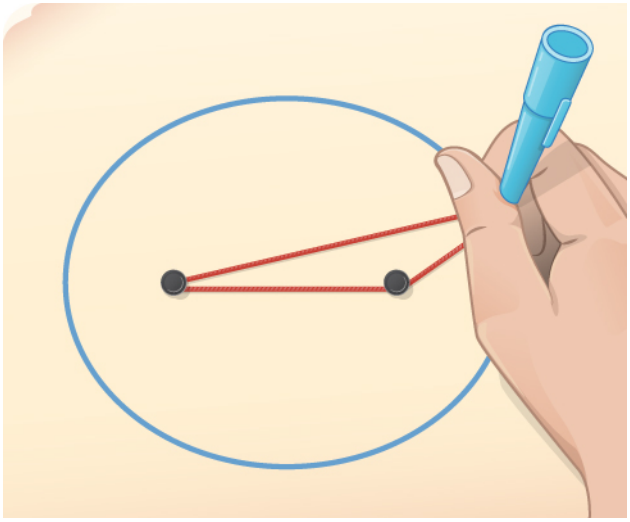


FIGURE 7.3 You can draw an ellipse as shown by putting a pin at each focus, and then placing a loop of string around a pen and the pins and tracing a line on the paper.

Kepler's Second Law

Each planet moves so that an imaginary line drawn from the sun to the planet sweeps out equal areas in equal times, as shown in Figure 7.4.

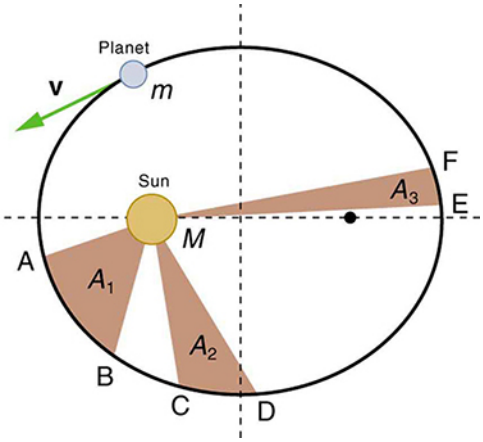


FIGURE 7.4 The shaded regions have equal areas. The time for *m* to go from A to B is the same as the time to go from C to D and from E to F. The mass *m* moves fastest when it is closest to *M*. Kepler's second law was originally devised for planets orbiting the sun, but it has broader validity.

TIPS FOR SUCCESS

Note that while, for historical reasons, Kepler's laws are stated for planets orbiting the sun, they are actually valid for all bodies satisfying the two previously stated conditions.

Kepler's Third Law

The ratio of the periods squared of any two planets around the sun is equal to the ratio of their average distances from the sun cubed. In equation form, this is

$$\frac{T_1^2}{T_2^2} = \frac{r_1^3}{r_2^3},$$

where T is the period (time for one orbit) and r is the average distance (also called orbital radius). This equation is valid only for comparing two small masses orbiting a single large mass. Most importantly, this is only a descriptive equation; it gives no information about the cause of the equality.

LINKS TO PHYSICS

History: Ptolemy vs. Copernicus

Before the discoveries of Kepler, Copernicus, Galileo, Newton, and others, the solar system was thought to revolve around Earth as shown in Figure 7.5 **(a)**. This is called the **Ptolemaic model**, named for the Greek philosopher Ptolemy who lived in the second century AD. The Ptolemaic model is characterized by a list of facts for the motions of planets, with no explanation of cause and effect. There tended to be a different rule for each heavenly body and a general lack of simplicity.

Figure 7.5 **(b)** represents the modern or **Copernican model**. In this model, a small set of rules and a single underlying force explain not only all planetary motion in the solar system, but also all other situations involving gravity. The breadth and simplicity of the laws of physics are compelling.

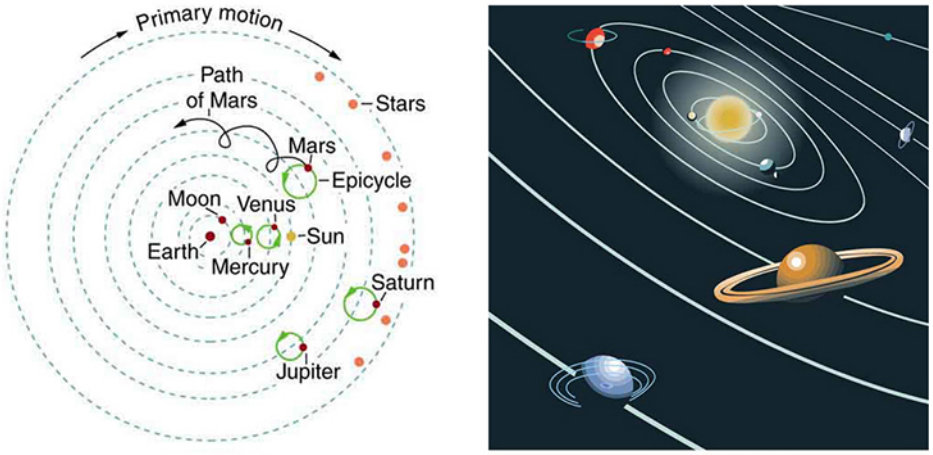


FIGURE 7.5 (a) The Ptolemaic model of the universe has Earth at the center with the moon, the planets, the sun, and the stars revolving about it in complex circular paths. This geocentric (Earth-centered) model, which can be made progressively more accurate by adding more circles, is purely descriptive, containing no hints about the causes of these motions. (b) The Copernican heliocentric (sun-centered) model is a simpler and more accurate model.

Nicolaus Copernicus (1473–1543) first had the idea that the planets circle the sun, in about 1514. It took him almost 20 years to work out the mathematical details for his model. He waited another 10 years or so to publish his work. It is thought he hesitated because he was afraid people would make fun of his theory. Actually, the reaction of many people was more one of fear and anger. Many people felt the Copernican model threatened their basic belief system. About 100 years later, the astronomer Galileo was put under house arrest for providing evidence that planets, including Earth, orbited the sun. In all, it took almost 300 years for everyone to admit that Copernicus had been right all along.

GRASP CHECK

Explain why Earth does actually appear to be the center of the solar system.

a. Earth appears to be the center of the solar system because Earth is at the center of the universe, and everything revolves around it in a circular orbit.
b. Earth appears to be the center of the solar system because, in the reference frame of Earth, the sun, moon, and planets all appear to move across the sky as if they were circling Earth.
c. Earth appears to be at the center of the solar system because Earth is at the center of the solar system and all the heavenly bodies revolve around it.
d. Earth appears to be at the center of the solar system because Earth is located at one of the foci of the elliptical orbit of the sun, moon, and other planets.

VIRTUAL PHYSICS

Acceleration

This simulation allows you to create your own solar system so that you can see how changing distances and masses determines the orbits of planets. Click *Help* for instructions.

Access multimedia content (http://openstax.org/books/physics/pages/7-1-keplers-laws-of-planetary-motion)

When the central object is off center, how does the speed of the orbiting object vary?

a. The orbiting object moves fastest when it is closest to the central object and slowest when it is farthest away.
b. The orbiting object moves slowest when it is closest to the central object and fastest when it is farthest away.
c. The orbiting object moves with the same speed at every point on the circumference of the elliptical orbit.
d. There is no relationship between the speed of the object and the location of the planet on the

circumference of the orbit.

Calculations Related to Kepler's Laws of Planetary Motion

Kepler's First Law

Refer back to Figure 7.2 **(a)**. Notice which distances are constant. The foci are fixed, so distance $\overline{f_1 f_2}$ is a constant. The definition of an ellipse states that the sum of the distances $\overline{f_1 m} + \overline{m f_2}$ is also constant. These two facts taken together mean that the perimeter of triangle $\Delta f_1 m f_2$ must also be constant. Knowledge of these constants will help you determine positions and distances of objects in a system that includes one object orbiting another.

Kepler's Second Law

Refer back to Figure 7.4. The second law says that the segments have equal area and that it takes equal time to sweep through each segment. That is, the time it takes to travel from A to B equals the time it takes to travel from C to D, and so forth. Velocity **v** equals distance *d* divided by time *t*: $\mathbf{v} = d/t$. Then, $t = d/\mathbf{v}$, so distance divided by velocity is also a constant. For example, if we know the average velocity of Earth on June 21 and December 21, we can compare the distance Earth travels on those days.

The degree of elongation of an elliptical orbit is called its **eccentricity** (*e*). Eccentricity is calculated by dividing the distance *f* from the center of an ellipse to one of the foci by half the long axis *a*.

$$(e) = f/a \tag{7.1}$$

When $e = 0$, the ellipse is a circle.

The area of an ellipse is given by $A = \pi ab$, where *b* is half the short axis. If you know the axes of Earth's orbit and the area Earth sweeps out in a given period of time, you can calculate the fraction of the year that has elapsed.

WORKED EXAMPLE

Kepler's First Law

At its closest approach, a moon comes within 200,000 km of the planet it orbits. At that point, the moon is 300,000 km from the other focus of its orbit, f_2. The planet is focus f_1 of the moon's elliptical orbit. How far is the moon from the planet when it is 260,000 km from f_2?

Strategy

Show and label the ellipse that is the orbit in your solution. Picture the triangle $f_1 m f_2$ collapsed along the major axis and add up the lengths of the three sides. Find the length of the unknown side of the triangle when the moon is 260,000 km from f_2.

Solution

Perimeter of $f_1 m f_2 = 200,000 \text{ km} + 100,000 \text{ km} + 300,000 \text{ km} = 600,000 \text{ km}.$

$mf_1 = 600,000 \text{ km} - (100,000 \text{ km} + 260,000 \text{ km}) = 240,000 \text{ km}.$

Discussion

The perimeter of triangle $f_1 m f_2$ must be constant because the distance between the foci does not change and Kepler's first law says the orbit is an ellipse. For any ellipse, the sum of the two sides of the triangle, which are $f_1 m$ and $m f_2$, is constant.

WORKED EXAMPLE

Kepler's Second Law

Figure 7.6 shows the major and minor axes of an ellipse. The semi-major and semi-minor axes are half of these, respectively.

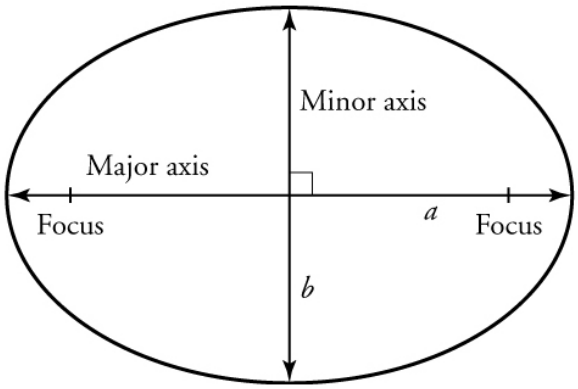


FIGURE 7.6 The major axis is the length of the ellipse, and the minor axis is the width of the ellipse. The semi-major axis is half the major axis, and the semi-minor axis is half the minor axis.

Earth's orbit is very slightly elliptical, with a semi-major axis of 1.49598×10^8 km and a semi-minor axis of 1.49577×10^8 km. If Earth's period is 365.26 days, what area does an Earth-to-sun line sweep past in one day?

Strategy

Each day, Earth sweeps past an equal-sized area, so we divide the total area by the number of days in a year to find the area swept past in one day. For total area use $A = \pi ab$. Calculate *A*, the area inside Earth's orbit and divide by the number of days in a year (i.e., its period).

Solution

$$\begin{aligned} \text{area per day} &= \frac{\text{total area}}{\text{total number of days}} \\ &= \frac{\pi ab}{365 \text{ d}} \\ &= \frac{\pi\left(1.496\times10^8 \text{ km}\right)\left(1.496\times10^3 \text{ km}\right)}{365 \text{ d}} \\ &= 1.93 \times 10^{14} \text{ km}^2/\text{d} \end{aligned} \qquad \textbf{7.2}$$

The area swept out in one day is thus $1.93 \times 10^{14}\text{km}^2$.

Discussion

The answer is based on Kepler's law, which states that a line from a planet to the sun sweeps out equal areas in equal times.

Kepler's Third Law

Kepler's third law states that the ratio of the squares of the periods of any two planets (T_1, T_2) is equal to the ratio of the cubes of their average orbital distance from the sun (r_1, r_2). Mathematically, this is represented by

$$\frac{T_1^2}{T_2^2} = \frac{r_1^3}{r_2^3}.$$

From this equation, it follows that the ratio r^3/T^2 is the same for all planets in the solar system. Later we will see how the work of Newton leads to a value for this constant.

WORKED EXAMPLE

Kepler's Third Law

Given that the moon orbits Earth each 27.3 days and that it is an average distance of 3.84×10^8 m from the center of Earth, calculate the period of an artificial satellite orbiting at an average altitude of 1,500 km above Earth's surface.

Strategy

The period, or time for one orbit, is related to the radius of the orbit by Kepler's third law, given in mathematical form by $\frac{T_1^2}{T_2^2} = \frac{r_1^3}{r_2^3}$. Let us use the subscript 1 for the moon and the subscript 2 for the satellite. We are asked to find T_2. The given information tells us that the orbital radius of the moon is $r_1 = 3.84 \times 10^8 \text{m}$, and that the period of the moon is $T_1 = 27.3 \text{ days}$. The height of the artificial satellite above Earth's surface is given, so to get the distance r_2 from the center of Earth we must add the height to the radius of Earth (6380 km). This gives $r_2 = 1500 \text{ km} + 6380 \text{ km} = 7880 \text{ km}$. Now all quantities are known, so T_2 can be found.

Solution

To solve for T_2, we cross-multiply and take the square root, yielding

$$T_2^2 = T_1^2 \left(\frac{r_2}{r_1}\right)^3; \; T_2 = T_1 \left(\frac{r_2}{r_1}\right)^{\frac{3}{2}} \tag{7.3}$$

$$T_2 = (27.3 \text{ d}) \left(\frac{24.0 \text{ h}}{\text{d}}\right) \left(\frac{7880 \text{ km}}{3.84 \times 10^5 \text{ km}}\right)^{\frac{3}{2}} = 1.93 \text{ h}.$$

Discussion

This is a reasonable period for a satellite in a fairly low orbit. It is interesting that any satellite at this altitude will complete one orbit in the same amount of time.

Practice Problems

1. A planet with no axial tilt is located in another solar system. It circles its sun in a very elliptical orbit so that the temperature varies greatly throughout the year. If the year there has 612 days and the inhabitants celebrate the coldest day on day 1 of their calendar, when is the warmest day?

a. Day 1
b. Day 153
c. Day 306
d. Day 459

2. A geosynchronous Earth satellite is one that has an orbital period of precisely 1 day. Such orbits are useful for communication and weather observation because the satellite remains above the same point on Earth (provided it orbits in the equatorial plane in the same direction as Earth's rotation). The ratio $\frac{r^3}{T^2}$ for the moon is $1.01 \times 10^{18} \frac{\text{km}^3}{y^2}$. Calculate the radius of the orbit of such a satellite.

a. $2.75 \times 10^3 \text{ km}$
b. $1.96 \times 10^4 \text{ km}$
c. $1.40 \times 10^5 \text{ km}$
d. $1.00 \times 10^6 \text{ km}$

Check Your Understanding

3. Are Kepler's laws purely descriptive, or do they contain causal information?

a. Kepler's laws are purely descriptive.
b. Kepler's laws are purely causal.
c. Kepler's laws are descriptive as well as causal.
d. Kepler's laws are neither descriptive nor causal.

4. True or false—According to Kepler's laws of planetary motion, a satellite increases its speed as it approaches its parent body and decreases its speed as it moves away from the parent body.

a. True

b. False

5. Identify the locations of the foci of an elliptical orbit.
   a. One focus is the parent body, and the other is located at the opposite end of the ellipse, at the same distance from the center as the parent body.
   b. One focus is the parent body, and the other is located at the opposite end of the ellipse, at half the distance from the center as the parent body.
   c. One focus is the parent body and the other is located outside of the elliptical orbit, on the line on which is the semi-major axis of the ellipse.
   d. One focus is on the line containing the semi-major axis of the ellipse, and the other is located anywhere on the elliptical orbit of the satellite.

7.2 Newton's Law of Universal Gravitation and Einstein's Theory of General Relativity

SECTION LEARNING OBJECTIVES

By the end of this section, you will be able to do the following:

- Explain Newton's law of universal gravitation and compare it to Einstein's theory of general relativity
- Perform calculations using Newton's law of universal gravitation

Section Key Terms

Einstein's theory of general relativity gravitational constant Newton's universal law of gravitation

Concepts Related to Newton's Law of Universal Gravitation

Sir Isaac Newton was the first scientist to precisely define the gravitational force, and to show that it could explain both falling bodies and astronomical motions. See Figure 7.7. But Newton was not the first to suspect that the same force caused both our weight and the motion of planets. His forerunner, Galileo Galilei, had contended that falling bodies and planetary motions had the same cause. Some of Newton's contemporaries, such as Robert Hooke, Christopher Wren, and Edmund Halley, had also made some progress toward understanding gravitation. But Newton was the first to propose an exact mathematical form and to use that form to show that the motion of heavenly bodies should be conic sections—circles, ellipses, parabolas, and hyperbolas. This theoretical prediction was a major triumph. It had been known for some time that moons, planets, and comets follow such paths, but no one had been able to propose an explanation of the mechanism that caused them to follow these paths and not others.



FIGURE 7.7 The popular legend that Newton suddenly discovered the law of universal gravitation when an apple fell from a tree and hit him

on the head has an element of truth in it. A more probable account is that he was walking through an orchard and wondered why all the apples fell in the same direction with the same acceleration. Great importance is attached to it because Newton's universal law of gravitation and his laws of motion answered very old questions about nature and gave tremendous support to the notion of underlying simplicity and unity in nature. Scientists still expect underlying simplicity to emerge from their ongoing inquiries into nature.

The gravitational force is relatively simple. It is always attractive, and it depends only on the masses involved and the distance between them. Expressed in modern language, **Newton's universal law of gravitation** states that every object in the universe attracts every other object with a force that is directed along a line joining them. The force is directly proportional to the product of their masses and inversely proportional to the square of the distance between them. This attraction is illustrated by Figure 7.8.

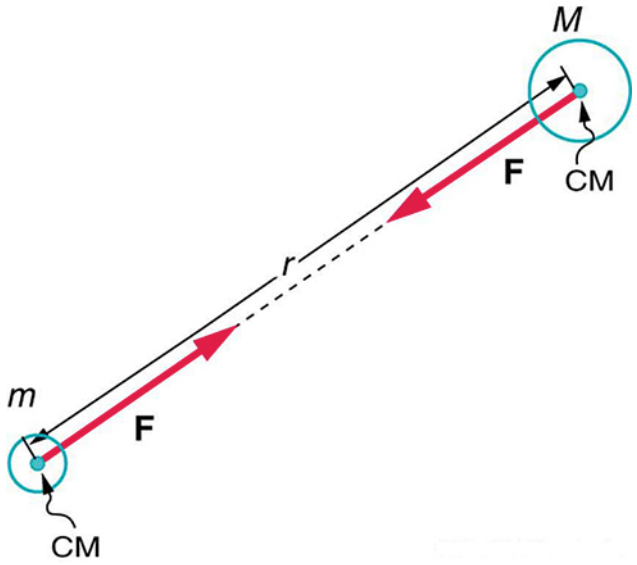


FIGURE 7.8 Gravitational attraction is along a line joining the centers of mass (CM) of the two bodies. The magnitude of the force on each body is the same, consistent with Newton's third law (action-reaction).

For two bodies having masses m and M with a distance r between their centers of mass, the equation for Newton's universal law of gravitation is

$$\mathbf{F} = G\frac{mM}{r^2}$$

where **F** is the magnitude of the gravitational force and G is a proportionality factor called the **gravitational constant**. G is a universal constant, meaning that it is thought to be the same everywhere in the universe. It has been measured experimentally to be $G = 6.673 \times 10^{-11}\,\text{N} \cdot \text{m}^2/\text{kg}^2$.

If a person has a mass of 60.0 kg, what would be the force of gravitational attraction on him at Earth's surface? G is given above, Earth's mass M is 5.97×10^{24} kg, and the radius r of Earth is 6.38×10^6 m. Putting these values into Newton's universal law of gravitation gives

$$\mathbf{F} = G\frac{mM}{r^2} = \left(6.673 \times 10^{-11}\,\frac{\text{N} \cdot \text{m}^2}{\text{kg}^2}\right)\left(\frac{(60.0\ \text{kg})\left(5.97 \times 10^{24}\ \text{kg}\right)}{\left(6.38 \times 10^6\ \text{m}\right)^2}\right) = 584\ \text{N}$$

We can check this result with the relationship: $\mathbf{F} = m\mathbf{g} = (60\ \text{kg})(9.8\text{m/s}^2) = 588\ \text{N}$

You may remember that **g**, the acceleration due to gravity, is another important constant related to gravity. By substituting **g** for **a** in the equation for Newton's second law of motion we get $\mathbf{F} = m\mathbf{g}$. Combining this with the equation for universal gravitation gives

$$m\mathbf{g} = G\frac{mM}{r^2}$$

Cancelling the mass m on both sides of the equation and filling in the values for the gravitational constant and mass and radius of the Earth, gives the value of g, which may look familiar.

$$\mathbf{g} = G\frac{M}{r^2} = \left(6.67 \times 10^{-11}\,\frac{\text{N} \cdot \text{m}^2}{\text{kg}^2}\right)\left(\frac{5.98 \times 10^{24}\text{kg}}{\left(6.38 \times 10^{6}\text{m}\right)^2}\right) = 9.80\text{m/s}^2$$

This is a good point to recall the difference between mass and weight. Mass is the amount of matter in an object; weight is the force of attraction between the mass within two objects. Weight can change because g is different on every moon and planet. An object's mass m does not change but its weight $m\mathbf{g}$ can.

VIRTUAL PHYSICS

Gravity and Orbits

Move the sun, Earth, moon and space station in this simulation (https://openstax.org/l/28gravityorbit) to see how it affects their gravitational forces and orbital paths. Visualize the sizes and distances between different heavenly bodies. Turn off gravity to see what would happen without it!

Access multimedia content (http://openstax.org/books/physics/pages/7-2-newtons-law-of-universal-gravitation-and-einsteins-theory-of-general-relativity)

Why doesn't the Moon travel in a smooth circle around the Sun?

a. The Moon is not affected by the gravitational field of the Sun.
b. The Moon is not affected by the gravitational field of the Earth.
c. The Moon is affected by the gravitational fields of both the Earth and the Sun, which are always additive.
d. The moon is affected by the gravitational fields of both the Earth and the Sun, which are sometimes additive and sometimes opposite.

SNAP LAB

Take-Home Experiment: Falling Objects

In this activity you will study the effects of mass and air resistance on the acceleration of falling objects. Make predictions (hypotheses) about the outcome of this experiment. Write them down to compare later with results.

- Four sheets of $8\text{-}1/2 \times 11$ -inch paper

Procedure

- Take four identical pieces of paper.
  - Crumple one up into a small ball.
  - Leave one uncrumpled.
  - Take the other two and crumple them up together, so that they make a ball of exactly twice the mass of the other crumpled ball.
  - Now compare which ball of paper lands first when dropped simultaneously from the same height.
    1. Compare crumpled one-paper ball with crumpled two-paper ball.
    2. Compare crumpled one-paper ball with uncrumpled paper.

Why do some objects fall faster than others near the surface of the Earth if the force of gravity is proportional to the mass?

a. Some objects fall faster because of air resistance, which acts in the direction of the motion of the object and exerts more force on objects with less surface area.
b. Some objects fall faster because of air resistance, which acts in the direction opposite the motion of the object and exerts more force on objects with less surface area.
c. Some objects fall faster because of air resistance, which acts in the direction of motion of the object and exerts more force on objects with more surface area.
d. Some objects fall faster because of air resistance, which acts in the direction opposite the motion of the object and exerts more force on objects with more surface area.

It is possible to derive Kepler's third law from Newton's law of universal gravitation. Applying Newton's second law

of motion to angular motion gives an expression for centripetal force, which can be equated to the expression for force in the universal gravitation equation. This expression can be manipulated to produce the equation for Kepler's third law. We saw earlier that the expression r^3/T^2 is a constant for satellites orbiting the same massive object. The derivation of Kepler's third law from Newton's law of universal gravitation and Newton's second law of motion yields that constant:

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

where M is the mass of the central body about which the satellites orbit (for example, the sun in our solar system). The usefulness of this equation will be seen later.

The universal gravitational constant G is determined experimentally. This determination was first done accurately in 1798 by English scientist Henry Cavendish (1731–1810), more than 100 years after Newton published his universal law of gravitation. The measurement of G is very basic and important because it determines the strength of one of the four forces in nature. Cavendish's experiment was very difficult because he measured the tiny gravitational attraction between two ordinary-sized masses (tens of kilograms at most) by using an apparatus like that in Figure 7.9. Remarkably, his value for G differs by less than 1% from the modern value.

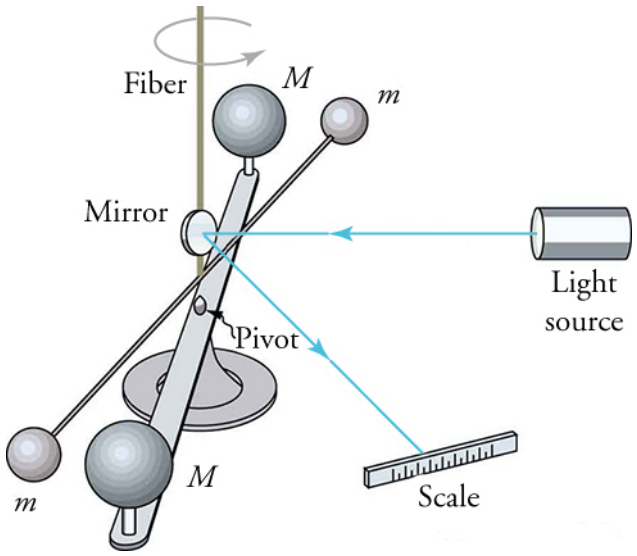


FIGURE 7.9 Cavendish used an apparatus like this to measure the gravitational attraction between two suspended spheres (m) and two spheres on a stand (M) by observing the amount of torsion (twisting) created in the fiber. The distance between the masses can be varied to check the dependence of the force on distance. Modern experiments of this type continue to explore gravity.

Einstein's Theory of General Relativity

Einstein's theory of general relativity explained some interesting properties of gravity not covered by Newton's theory. Einstein based his theory on the postulate that acceleration and gravity have the same effect and cannot be distinguished from each other. He concluded that light must fall in both a gravitational field and in an accelerating reference frame. Figure 7.10 shows this effect (greatly exaggerated) in an accelerating elevator. In Figure 7.10 **(a)**, the elevator accelerates upward in zero gravity. In Figure 7.10 **(b)**, the room is not accelerating but is subject to gravity. The effect on light is the same: it "falls" downward in both situations. The person in the elevator cannot tell whether the elevator is accelerating in zero gravity or is stationary and subject to gravity. Thus, gravity affects the path of light, even though we think of gravity as acting between masses, while photons are massless.

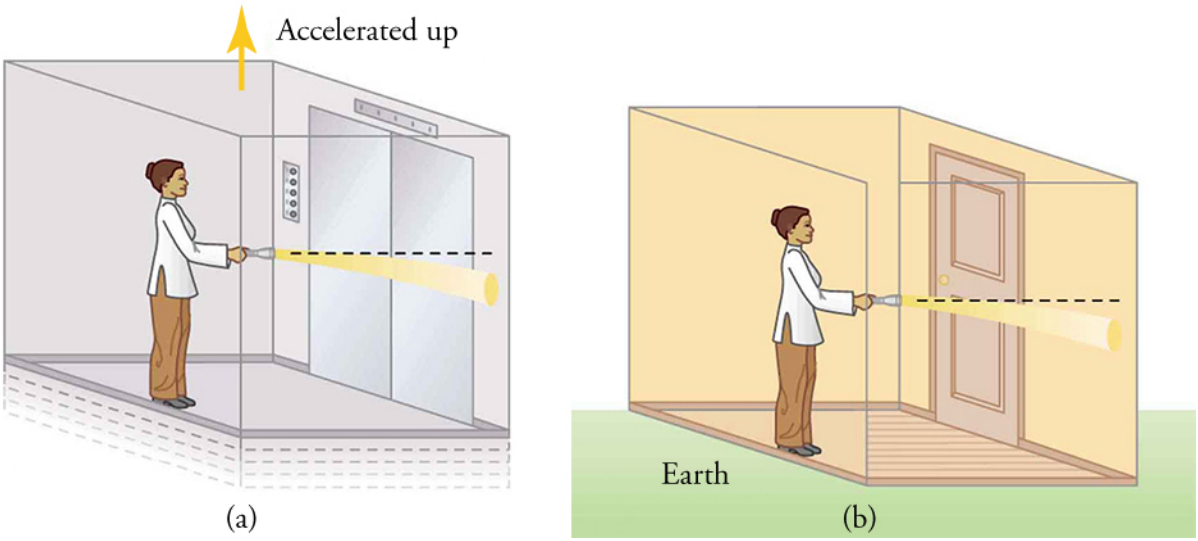


FIGURE 7.10 (a) A beam of light emerges from a flashlight in an upward-accelerating elevator. Since the elevator moves up during the time the light takes to reach the wall, the beam strikes lower than it would if the elevator were not accelerated. (b) Gravity must have the same effect on light, since it is not possible to tell whether the elevator is accelerating upward or is stationary and acted upon by gravity.

Einstein's theory of general relativity got its first verification in 1919 when starlight passing near the sun was observed during a solar eclipse. (See Figure 7.11.) During an eclipse, the sky is darkened and we can briefly see stars. Those on a line of sight nearest the sun should have a shift in their apparent positions. Not only was this shift observed, but it agreed with Einstein's predictions well within experimental uncertainties. This discovery created a scientific and public sensation. Einstein was now a folk hero as well as a very great scientist. The bending of light by matter is equivalent to a bending of space itself, with light following the curve. This is another radical change in our concept of space and time. It is also another connection that any particle with mass *or* energy (e.g., massless photons) is affected by gravity.

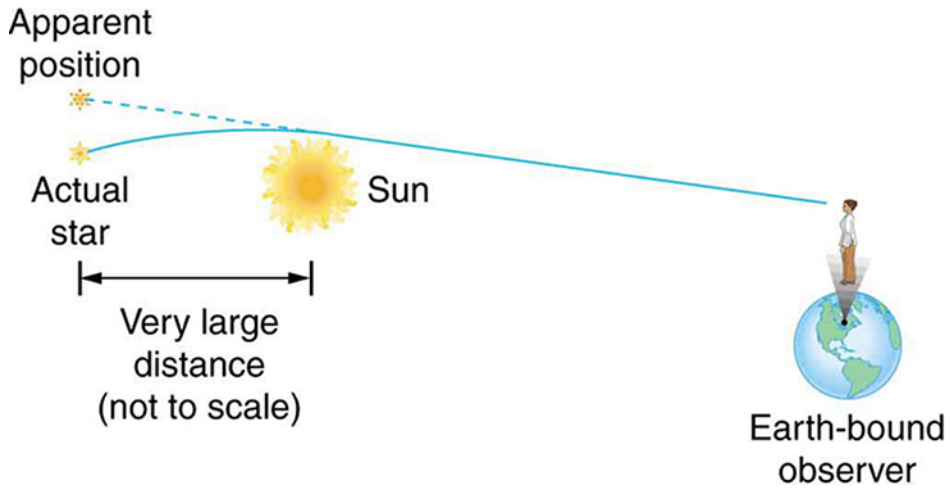


FIGURE 7.11 This schematic shows how light passing near a massive body like the sun is curved toward it. The light that reaches the Earth then seems to be coming from different locations than the known positions of the originating stars. Not only was this effect observed, but the amount of bending was precisely what Einstein predicted in his general theory of relativity.

To summarize the two views of gravity, Newton envisioned gravity as a tug of war along the line connecting any two objects in the universe. In contrast, Einstein envisioned gravity as a bending of space-time by mass.

BOUNDLESS PHYSICS

NASA gravity probe B

NASA's Gravity Probe B (GP-B) mission has confirmed two key predictions derived from Albert Einstein's general

theory of relativity. The probe, shown in Figure 7.12 was launched in 2004. It carried four ultra-precise gyroscopes designed to measure two effects hypothesized by Einstein's theory:

- The geodetic effect, which is the warping of space and time by the gravitational field of a massive body (in this case, Earth)
- The frame-dragging effect, which is the amount by which a spinning object pulls space and time with it as it rotates



FIGURE 7.12 Artist concept of Gravity Probe B spacecraft in orbit around the Earth. (credit: NASA/MSFC)

Both effects were measured with unprecedented precision. This was done by pointing the gyroscopes at a single star while orbiting Earth in a polar orbit. As predicted by relativity theory, the gyroscopes experienced very small, but measureable, changes in the direction of their spin caused by the pull of Earth's gravity.

The principle investigator suggested imagining Earth spinning in honey. As Earth rotates it drags space and time with it as it would a surrounding sea of honey.

According to the general theory of relativity, a gravitational field bends light. What does this have to do with time and space?

a. Gravity has no effect on the space-time continuum, and gravity only affects the motion of light.
b. The space-time continuum is distorted by gravity, and gravity has no effect on the motion of light.
c. Gravity has no effect on either the space-time continuum or on the motion of light.
d. The space-time continuum is distorted by gravity, and gravity affects the motion of light.

Calculations Based on Newton's Law of Universal Gravitation

TIPS FOR SUCCESS

When performing calculations using the equations in this chapter, use units of kilograms for mass, meters for distances, newtons for force, and seconds for time.

The mass of an object is constant, but its weight varies with the strength of the gravitational field. This means the value of $\mathbf{g}$ varies from place to place in the universe. The relationship between force, mass, and acceleration from the second law of motion can be written in terms of $\mathbf{g}$.

$$\mathbf{F} = m\mathbf{a} = m\mathbf{g}$$

In this case, the force is the weight of the object, which is caused by the gravitational attraction of the planet or moon on which the object is located. We can use this expression to compare weights of an object on different moons and planets.

WATCH PHYSICS

Mass and Weight Clarification

This video shows the mathematical basis of the relationship between mass and weight. The distinction between mass and weight are clearly explained. The mathematical relationship between mass and weight are shown mathematically in terms of the equation for Newton's law of universal gravitation and in terms of his second law of motion.

Mass and weight clarification | Centripetal force and gravitation | Physics | Khan Academy (https://openstax.org/l/28massweight)

GRASP CHECK

Would you have the same mass on the moon as you do on Earth? Would you have the same weight?

a. You would weigh more on the moon than on Earth because gravity on the moon is stronger than gravity on Earth.
b. You would weigh less on the moon than on Earth because gravity on the moon is weaker than gravity on Earth.
c. You would weigh less on the moon than on Earth because gravity on the moon is stronger than gravity on Earth.
d. You would weigh more on the moon than on Earth because gravity on the moon is weaker than gravity on Earth.

Two equations involving the gravitational constant, *G*, are often useful. The first is Newton's equation, $\mathbf{F} = G\frac{mM}{r^2}$. Several of the values in this equation are either constants or easily obtainable. **F** is often the weight of an object on the surface of a large object with mass *M*, which is usually known. The mass of the smaller object, *m*, is often known, and *G* is a universal constant with the same value anywhere in the universe. This equation can be used to solve problems involving an object on or orbiting Earth or other massive celestial object. Sometimes it is helpful to equate the right-hand side of the equation to $m\mathbf{g}$ and cancel the *m* on both sides.

The equation $\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$ is also useful for problems involving objects in orbit. Note that there is no need to know the mass of the object. Often, we know the radius *r* or the period *T* and want to find the other. If these are both known, we can use the equation to calculate the mass of a planet or star.

WATCH PHYSICS

Mass and Weight Clarification

This video demonstrates calculations involving Newton's universal law of gravitation.

Introduction to Newton's law of gravitation | Physics | Khan Academy (https://openstax.org/l/02newlawgrav)

What is the difference between *g* and *G*?

a. *g* and *G* are both unchanging constants but have different units.
b. *G* is a universal constant that relates force with a pair of masses at a distance, while *g* relates force with mass and varies with location.
c. *g* describes acceleration while *G* describes gravitational force.
d. *g* describes gravitational force while *G* describes acceleration.

WORKED EXAMPLE

Change in g

The value of **g** on the planet Mars is 3.71 m/s^2. If you have a mass of 60.0 kg on Earth, what would be your mass on Mars? What would be your weight on Mars?

Strategy

Weight equals acceleration due to gravity times mass: $\mathbf{W} = m\mathbf{g}$. An object's mass is constant. Call acceleration due to gravity on Mars $\mathbf{g}_M$ and weight on Mars $\mathbf{W}_M$.

Solution

Mass on Mars would be the same, 60 kg.

$$\mathbf{W}_M = m\mathbf{g}_M = (60.0\text{ kg})\left(3.71\text{m/s}^2\right) = 223\text{ N} \quad \textbf{7.4}$$

Discussion

The value of **g** on any planet depends on the mass of the planet and the distance from its center. If the material below the surface varies from point to point, the value of **g** will also vary slightly.

WORKED EXAMPLE

Earth's g at the Moon

Find the acceleration due to Earth's gravity at the distance of the moon.

$$G = 6.67 \times 10^{-11}\text{ N}\cdot\text{m}^2/\text{kg}^2$$
$$\text{Earth-moon distance} = 3.84 \times 10^8\text{ m} \quad \textbf{7.5}$$
$$\text{Earth's mass} = 5.98 \times 10^{24}\text{ kg}$$

Express the force of gravity in terms of *g*.

$$\text{F} = \text{W} = m\text{a} = m\text{g} \quad \textbf{7.6}$$

Combine with the equation for universal gravitation.

$$m\mathbf{g} = mG\frac{M}{r^2} \quad \textbf{7.7}$$

Solution

Cancel *m* and substitute.

$$\mathbf{g} = G\frac{M}{r^2} = \left(6.67\times10^{-11}\frac{\text{N}\cdot\text{m}^2}{\text{kg}^2}\right)\left(\frac{5.98\times10^{24}\text{ kg}}{\left(3.84\times10^8\text{ m}\right)^2}\right) = 2.70\times10^{-3}\text{m/s}^2 \quad \textbf{7.8}$$

Discussion

The value of **g** for the moon is 1.62 m/s^2. Comparing this value to the answer, we see that Earth's gravitational influence on an object on the moon's surface would be insignificant.

Practice Problems

6. What is the mass of a person who weighs 600 N?

a. 6.00 kg
b. 61.2 kg
c. 600 kg
d. 610 kg

7. Calculate Earth's mass given that the acceleration due to gravity at the North Pole is 9.830m/s^2 and the radius of the Earth is 6371 km from pole to center.

a. 5.94×10^{17} kg
b. 5.94×10^{24}kg
c. 9.36×10^{17}kg
d. 9.36×10^{24}kg

Check Your Understanding

8. Some of Newton's predecessors and contemporaries also studied gravity and proposed theories. What important advance did Newton make in the study of gravity that the other scientists had failed to do?

a. He gave an exact mathematical form for the theory.
b. He added a correction term to a previously existing formula.

c. Newton found the value of the universal gravitational constant.
d. Newton showed that gravitational force is always attractive.

9. State the law of universal gravitation in words only.
a. Gravitational force between two objects is directly proportional to the sum of the squares of their masses and inversely proportional to the square of the distance between them.
b. Gravitational force between two objects is directly proportional to the product of their masses and inversely proportional to the square of the distance between them.
c. Gravitational force between two objects is directly proportional to the sum of the squares of their masses and inversely proportional to the distance between them.
d. Gravitational force between two objects is directly proportional to the product of their masses and inversely proportional to the distance between them.

10. Newton’s law of universal gravitation explains the paths of what?
a. A charged particle
b. A ball rolling on a plane surface
c. A planet moving around the sun
d. A stone tied to a string and whirled at constant speed in a horizontal circle

Key Terms

aphelion the point farthest from the sun in the path of an orbiting planet (called apoapsis for other celestial bodies)

Copernican model the model of the solar system where the sun is at the center of the solar system and all the planets orbit around it; this is also called the heliocentric model

eccentricity a measure of the separation of the foci of an ellipse

Einstein's theory of general relativity the theory that gravitational force results from the bending of spacetime by an object's mass

gravitational constant the proportionality constant in Newton's law of universal gravitation

Kepler's laws of planetary motion three laws derived by Johannes Kepler that describe the properties of all orbiting satellites

Newton's universal law of gravitation states that gravitational force between two objects is directly proportional to the product of their masses and inversely proportional to the square of the distance between them.

perihelion closest distance between a planet and the sun (called periapsis for other celestial bodies)

Ptolemaic model the model of the solar system where Earth is at the center of the solar system and the sun and all the planets orbit around it; this is also called the geocentric model

Section Summary

7.1 Kepler's Laws of Planetary Motion

- All satellites follow elliptical orbits.
- The line from the satellite to the parent body sweeps out equal areas in equal time.
- The radius cubed divided by the period squared is a constant for all satellites orbiting the same parent body.

7.2 Newton's Law of Universal Gravitation and Einstein's Theory of General Relativity

- Newton's law of universal gravitation provides a mathematical basis for gravitational force and Kepler's laws of planetary motion.
- Einstein's theory of general relativity shows that gravitational fields change the path of light and warp space and time.
- An object's mass is constant, but its weight changes when acceleration due to gravity, **g**, changes.

Key Equations

7.1 Kepler's Laws of Planetary Motion

Kepler's third law	$\frac{T_1^2}{T_2^2} = \frac{r_1^3}{r_2^3}$
eccentricity	$e = \frac{f}{a}$
area of an ellipse	$A = \pi ab$
semi-major axis of an ellipse	$a = \left(r_\text{a} + r_\text{p}\right)/2$
semi-minor axis of an ellipse	$b = \sqrt{r_\text{a} r_\text{p}}$

7.2 Newton's Law of Universal Gravitation and Einstein's Theory of General Relativity

Newton's second law of motion	$\mathbf{F} = m\mathbf{a} = m\mathbf{g}$
Newton's universal law of gravitation	$\mathbf{F} = G\frac{mM}{r^2}$
acceleration due to gravity	$\mathbf{g} = G\frac{M}{r^2}$
constant for satellites orbiting the same massive object	$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$

Chapter Review

Concept Items

7.1 Kepler's Laws of Planetary Motion

1. A circle is a special case of an ellipse. Explain how a circle is different from other ellipses.

a. The foci of a circle are at the same point and are located at the center of the circle.

b. The foci of a circle are at the same point and are located at the circumference of the circle.

c. The foci of a circle are at the same point and are located outside of the circle.

d. The foci of a circle are at the same point and are located anywhere on the diameter, except on its midpoint.

2. Comets have very elongated elliptical orbits with the sun at one focus. Using Kepler's Law, explain why a comet travels much faster near the sun

than it does at the other end of the orbit.
   a. Because the satellite sweeps out equal areas in equal times
   b. Because the satellite sweeps out unequal areas in equal times
   c. Because the satellite is at the other focus of the ellipse
   d. Because the square of the period of the satellite is proportional to the cube of its average distance from the sun

3. True or False—A planet-satellite system must be isolated from other massive objects to follow Kepler's laws of planetary motion.
   a. True
   b. False

4. Explain why the string, pins, and pencil method works for drawing an ellipse.
   a. The string, pins, and pencil method works because the length of the two sides of the triangle remains constant as you are drawing the ellipse.
   b. The string, pins, and pencil method works because the area of the triangle remains constant as you are drawing the ellipse.
   c. The string, pins, and pencil method works because the perimeter of the triangle remains constant as you are drawing the ellipse.
   d. The string, pins, and pencil method works because the volume of the triangle remains constant as you are drawing the ellipse.

7.2 Newton's Law of Universal Gravitation and Einstein's Theory of General Relativity

5. Describe the postulate on which Einstein based the theory of general relativity and describe an everyday experience that illustrates this postulate.
   a. Gravity and velocity have the same effect and cannot be distinguished from each other. An acceptable illustration of this is any description of the feeling of constant velocity in a situation where no outside frame of reference is considered.
   b. Gravity and velocity have different effects and can be distinguished from each other. An acceptable illustration of this is any description of the feeling of constant velocity in a situation where no outside frame of reference is considered.
   c. Gravity and acceleration have the same effect and cannot be distinguished from each other. An acceptable illustration of this is any description of the feeling of acceleration in a situation where no outside frame of reference is considered.
   d. Gravity and acceleration have different effects and can be distinguished from each other. An acceptable illustration of this is any description of the feeling of acceleration in a situation where no outside frame of reference is considered.

6. Titan, with a radius of 2.58×10^6 m, is the largest moon of the planet Saturn. If the mass of Titan is 1.35×10^{23} kg, what is the acceleration due to gravity on the surface of this moon?
   a. $1.35\ \text{m/s}^2$
   b. $3.49\ \text{m/s}^2$
   c. $3.49 \times 10^6\ \text{m/s}^2$
   d. $1.35 \times 10^6\ \text{m/s}^2$

7. Saturn's moon Titan has an orbital period of 15.9 days. If Saturn has a mass of 5.68×10^{26} kg, what is the average distance from Titan to the center of Saturn?
   a. 1.22×10^9 m
   b. 4.26×10^7 m
   c. 5.25×10^4 km
   d. 4.26×10^{10} km

8. Explain why doubling the mass of an object doubles its weight, but doubling its distance from the center of Earth reduces its weight fourfold.
   a. The weight is two times the gravitational force between the object and Earth.
   b. The weight is half the gravitational force between the object and Earth.
   c. The weight is equal to the gravitational force between the object and Earth, and the gravitational force is inversely proportional to the distance squared between the object and Earth.
   d. The weight is directly proportional to the square of the gravitational force between the object and Earth.

9. Explain why a star on the other side of the Sun might appear to be in a location that is not its true location.

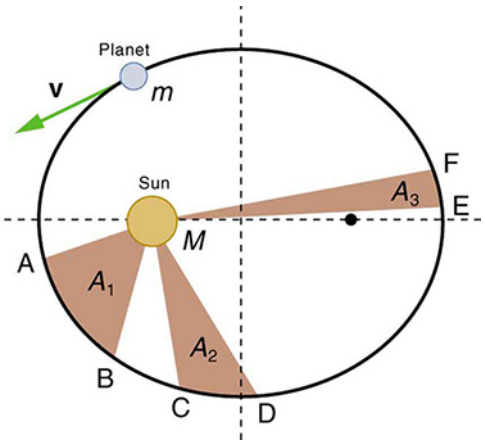
a. It can be explained by using the concept of atmospheric refraction.
b. It can be explained by using the concept of the special theory of relativity.
c. It can be explained by using the concept of the general theory of relativity.
d. It can be explained by using the concept of light scattering in the atmosphere.

10. The Cavendish experiment marked a milestone in the study of gravity.
Part A. What important value did the experiment determine?
Part B. Why was this so difficult in terms of the masses used in the apparatus and the strength of the gravitational force?
a. Part A. The experiment measured the acceleration due to gravity, g. Part B. Gravity is a very weak force but despite this limitation, Cavendish was able to measure the attraction between very massive objects.
b. Part A. The experiment measured the gravitational constant, G. Part B. Gravity is a very weak force but, despite this limitation, Cavendish was able to measure the attraction between very massive objects.
c. Part A. The experiment measured the acceleration due to gravity, g. Part B. Gravity is a very weak force but despite this limitation, Cavendish was able to measure the attraction between less massive objects.
d. Part A. The experiment measured the gravitational constant, G. Part B. Gravity is a very weak force but despite this limitation, Cavendish was able to measure the attraction between less massive objects.

Critical Thinking Items

7.1 Kepler's Laws of Planetary Motion

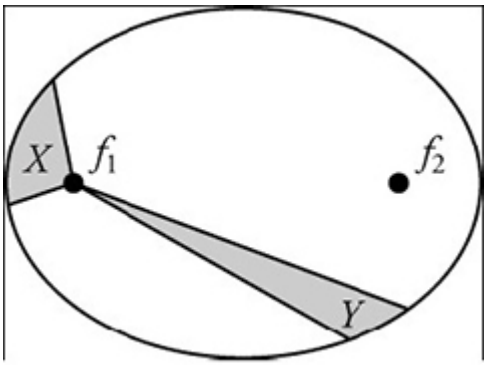
11. In the figure, the time it takes for the planet to go from A to B, C to D, and E to F is the same.



Compare the areas A_1, A_2, and A_3 in terms of size.
a. $A_1 \neq A_2 \neq A_3$
b. $A_1 = A_2 = A_3$
c. $A_1 = A_2 > A_3$
d. $A_1 > A_2 = A_3$

12. A moon orbits a planet in an elliptical orbit. The foci of the ellipse are 50, 000 km apart. The closest approach of the moon to the planet is 400, 000 km. What is the length of the major axis of the orbit?
a. 400, 000 km
b. 450, 000, km
c. 800, 000 km
d. 850, 000 km

13. In this figure, if f_1 represents the parent body, which set of statements holds true?



a. Area X < Area Y; the speed is greater for area X.
b. Area X > Area Y; the speed is greater for area Y.
c. Area X = Area Y; the speed is greater for area X.
d. Area X = Area Y; the speed is greater for area Y.

7.2 Newton's Law of Universal Gravitation and Einstein's Theory of General Relativity

14. Rhea, with a radius of 7.63×10^5 m, is the

second-largest moon of the planet Saturn. If the mass of Rhea is 2.31×10^{21} kg, what is the acceleration due to gravity on the surface of this moon?

a. 2.65×10^{-1} m/s
b. 2.02×10^{5} m/s
c. 2.65×10^{-1} m/s^2
d. 2.02×10^{5} m/s^2

15. Earth has a mass of 5.971×10^{24} kg and a radius of 6.371×10^{6} m. Use the data to check the value of the gravitational constant.

a. $6.66\times10^{-11}\ \frac{\text{N}\cdot\text{m}}{\text{kg}^2}$, it matches the value of the gravitational constant G.
b. $1.05\times10^{-17}\ \frac{\text{N}\cdot\text{m}}{\text{kg}^2}$, it matches the value of the gravitational constant G.
c. $6.66\times10^{-11}\ \frac{\text{N}\cdot\text{m}^2}{\text{kg}^2}$, it matches the value of the gravitational constant G.
d. $1.05\times10^{-17}\ \frac{\text{N}\cdot\text{m}^2}{\text{kg}^2}$, it matches the value of the gravitational constant G.

16. The orbit of the planet Mercury has a period of 88.0 days and an average radius of 5.791×10^{10} m. What is the mass of the sun?

a. 3.43×10^{19} kg
b. 1.99×10^{30} kg
c. 2.56×10^{29} kg
d. 1.48×10^{40} kg

Problems

7.1 Kepler's Laws of Planetary Motion

17. The closest Earth comes to the sun is 1.47×10^{8} km, and Earth's farthest distance from the sun is 1.52×10^{8} km. What is the area inside Earth's orbit?

a. 2.23×10^{16} km^2
b. 6.79×10^{16} km^2
c. 7.02×10^{16} km^2
d. 7.26×10^{16} km^2

18. Earth is 1.496×10^{8} km from the sun, and Neptune is 4.490×10^{9} km from the sun. What best represents the number of Earth years it takes for Neptune to complete one orbit around the sun?

a. 10 years
b. 30 years
c. 160 years
d. 900 years

Performance Task

7.2 Newton's Law of Universal Gravitation and Einstein's Theory of General Relativity

19. Design an experiment to test whether magnetic force is inversely proportional to the square of distance. Gravitational, magnetic, and electrical fields all act at a distance, but do they all follow the inverse square law? One difference in the forces related to these fields is that gravity is only attractive, but the other two can repel as well. In general, the inverse square law says that force *F* equals a constant *C* divided by the distance between objects, *d*, squared: $F = C/d^2$.

Incorporate these materials into your design:

- Two strong, permanent bar magnets
- A spring scale that can measure small forces
- A short ruler calibrated in millimeters

Use the magnets to study the relationship between attractive force and distance.

a. What will be the independent variable?
b. What will be the dependent variable?
c. How will you measure each of these variables?
d. If you plot the independent variable versus the dependent variable and the inverse square law is upheld, will the plot be a straight line? Explain.
e. Which plot would be a straight line if the inverse square law were upheld?

Test Prep

Multiple Choice

7.1 Kepler's Laws of Planetary Motion

20. A planet of mass m circles a sun of mass M. Which distance changes throughout the planet's orbit?

a. $\overline{f_1 f_2}$

b. $\overline{mM}$
c. $\overline{Mf_2}$
d. $\overline{Mf_1}$

21. The focal point of the elliptical orbit of a moon is $50,000\,\text{km}$ from the center of the orbit. If the eccentricity of the orbit is 0.25, what is the length of the semi-major axis?
a. $12,500\,\text{km}$
b. $100,000\,\text{km}$
c. $200,000\,\text{km}$
d. $400,000\,\text{km}$

22. An artificial satellite orbits the Earth at a distance of 1.45×10^4 km from Earth's center. The moon orbits the Earth at a distance of 3.84×10^5 km once every 27.3 days. How long does it take the satellite to orbit the Earth?
a. 0.200 days
b. 3.07 days
c. 243 days
d. 3721 days

23. Earth is 1.496×10^8 km from the sun, and Venus is 1.08×10^8 km from the sun. One day on Venus is 243 Earth days long. What best represents the number of Venusian days in a Venusian year?
a. 0.78 days
b. 0.92 days
c. 1.08 days
d. 1.21 days

7.2 Newton's Law of Universal Gravitation and Einstein's Theory of General Relativity

24. What did the Cavendish experiment measure?
a. The mass of Earth
b. The gravitational constant
c. Acceleration due to gravity
d. The eccentricity of Earth's orbit

25. You have a mass of 55 kg as measured by your bathroom scale, which is calibrated for Earth's gravity. You take the scale with you from Earth to a moon of Jupiter. When you weigh yourself at your destination, you appear to weigh only 6.9 kg. What is the acceleration due to gravity on this moon?
a. 0.810 m/s^2
b. 1.23 m/s^2
c. 5.39 m/s^2
d. 37.4 m/s^2

26. A person is in an elevator that suddenly begins to descend. The person knows, intuitively, that the feeling of suddenly becoming lighter is because the elevator is accelerating downward. What other change would produce the same feeling? How does this demonstrate Einstein's postulate on which he based the theory of general relativity?
a. It would feel the same if the force of gravity suddenly became weaker. This illustrates Einstein's postulates that gravity and acceleration are indistinguishable.
b. It would feel the same if the force of gravity suddenly became stronger. This illustrates Einstein's postulates that gravity and acceleration are indistinguishable.
c. It would feel the same if the force of gravity suddenly became weaker. This illustrates Einstein's postulates that gravity and acceleration are distinguishable.
d. It would feel the same if the force of gravity suddenly became stronger. This illustrates Einstein's postulates that gravity and acceleration are distinguishable.

Short Answer

7.1 Kepler's Laws of Planetary Motion

27. Explain how the masses of a satellite and its parent body must compare in order to apply Kepler's laws of planetary motion.
a. The mass of the parent body must be much less than that of the satellite.
b. The mass of the parent body must be much greater than that of the satellite.
c. The mass of the parent body must be equal to the mass of the satellite.
d. There is no specific relationship between the masses for applying Kepler's laws of planetary motion.

28. Hyperion is a moon of the planet Saturn. Its orbit has an eccentricity of 0.123 and a semi-major axis of 1.48×10^6 km. How far is the center of the orbit from the center of Saturn?
a. 1.82×10^5 km
b. 3.64×10^5 km

c. 1.20×10^7 km
d. 2.41×10^7 km

29. The orbits of satellites are elliptical. Define an ellipse.
a. An ellipse is an open curve wherein the sum of the distance from the foci to any point on the curve is constant.
b. An ellipse is a closed curve wherein the sum of the distance from the foci to any point on the curve is constant.
c. An ellipse is an open curve wherein the distances from the two foci to any point on the curve are equal.
d. An ellipse is a closed curve wherein the distances from the two foci to any point on the curve are equal.

30. The planet Mars has two moons, Deimos and Phobos. Deimos orbits Mars every 1.26 days with an average radius of 2.35×10^3 km. Phobos has an average orbit about four times greater in radius than Deimos. How long does it take for Phobos to make an orbit of Mars?
a. 2.24 days
b. 5.04 days
c. 9.40 days
d. 10.0 days

7.2 Newton's Law of Universal Gravitation and Einstein's Theory of General Relativity

31. Newton's third law of motion says that, for every action force, there is a reaction force equal in magnitude but that acts in the opposite direction. Apply this law to gravitational forces acting between the Washington Monument and Earth.
a. The monument is attracted to Earth with a force equal to its weight, and Earth is attracted to the monument with a force equal to Earth's weight. The situation can be represented with two force vectors of unequal magnitude and pointing in the same direction.
b. The monument is attracted to Earth with a force equal to its weight, and Earth is attracted to the monument with a force equal to Earth's weight. The situation can be represented with two force vectors of unequal magnitude but pointing in opposite directions.
c. The monument is attracted to Earth with a force equal to its weight, and Earth is attracted to the monument with an equal force. The situation can be represented with two force vectors of equal magnitude and pointing in the same direction.
d. The monument is attracted to Earth with a force equal to its weight, and Earth is attracted to the monument with an equal force. The situation can be represented with two force vectors of equal magnitude but pointing in opposite directions.

32. True or false—Gravitational force is the attraction of the mass of one object to the mass of another. Light, either as a particle or a wave, has no rest mass. Despite this fact gravity bends a beam of light.
a. True
b. False

33. The average radius of Earth is 6.37×10^6 m. What is Earth's mass?
a. 9.35×10^{17} kg
b. 5.96×10^{24} kg
c. 3.79×10^{31} kg
d. 2.42×10^{38} kg

34. What is the gravitational force between two 60.0 kg people sitting 100 m apart?
a. 2.4×10^{-11} N
b. 2.4×10^{-9} N
c. 3.6×10^{-1} N
d. 3.6×10^{1} N

Extended Response

7.1 Kepler's Laws of Planetary Motion

35. The orbit of Halley''s Comet has an eccentricity of 0.967 and stretches to the edge of the solar system.
Part A. Describe the shape of the comet's orbit.
Part B. Compare the distance traveled per day when it is near the sun to the distance traveled per day when it is at the edge of the solar system.
Part C. Describe variations in the comet's speed as it completes an orbit. Explain the variations in terms of Kepler's second law of planetary motion.

a. Part A. The orbit is circular, with the sun at the center. Part B. The comet travels much farther when it is near the sun than when it is at the edge of the solar system. Part C. The comet decelerates as it approaches the sun and accelerates as it leaves the sun.
b. Part A. The orbit is circular, with the sun at the center. Part B. The comet travels much farther when it is near the sun than when it is at the edge of the solar system. Part C. The comet accelerates as it approaches the sun and decelerates as it leaves the sun.
c. Part A. The orbit is very elongated, with the sun near one end. Part B. The comet travels much farther when it is near the sun than when it is at the edge of the solar system. Part C. The comet decelerates as it approaches the sun and accelerates as it moves away from the sun.

36. For convenience, astronomers often use astronomical units (AU) to measure distances within the solar system. One AU equals the average distance from Earth to the sun. Halley's Comet returns once every 75.3 years. What is the average radius of the orbit of Halley's Comet in AU?
a. 0.002 AU
b. 0.056 AU
c. 17.8 AU
d. 653 AU

7.2 Newton's Law of Universal Gravitation and Einstein's Theory of General Relativity

37. It took scientists a long time to arrive at the understanding of gravity as explained by Galileo and Newton. They were hindered by two ideas that seemed like common sense but were serious misconceptions. First was the fact that heavier things fall faster than light things. Second, it was believed impossible that forces could act at a distance. Explain why these ideas persisted and why they prevented advances.
a. Heavier things fall faster than light things if they have less surface area and greater mass density. In the Renaissance and before, forces that acted at a distance were considered impossible, so people were skeptical about scientific theories that invoked such forces.
b. Heavier things fall faster than light things because they have greater surface area and less mass density. In the Renaissance and before, forces that act at a distance were considered impossible, so people were skeptical about scientific theories that invoked such forces.
c. Heavier things fall faster than light things because they have less surface area and greater mass density. In the Renaissance and before, forces that act at a distance were considered impossible, so people were quick to accept scientific theories that invoked such forces.
d. Heavier things fall faster than light things because they have larger surface area and less mass density. In the Renaissance and before, forces that act at a distance were considered impossible because of people's faith in scientific theories.

38. The masses of Earth and the moon are 5.97×10^{24} kg and 7.35×10^{22} kg, respectively. The distance from Earth to the moon is 3.80×10^{5} km. At what point between the Earth and the moon are the opposing gravitational forces equal? (Use subscripts e and m to represent Earth and moon.)
a. 3.42×10^{5} km from the center of Earth
b. 3.80×10^{5} km from the center of Earth
c. 3.42×10^{6} km from the center of Earth
d. 3.10×10^{7} km from the center of Earth